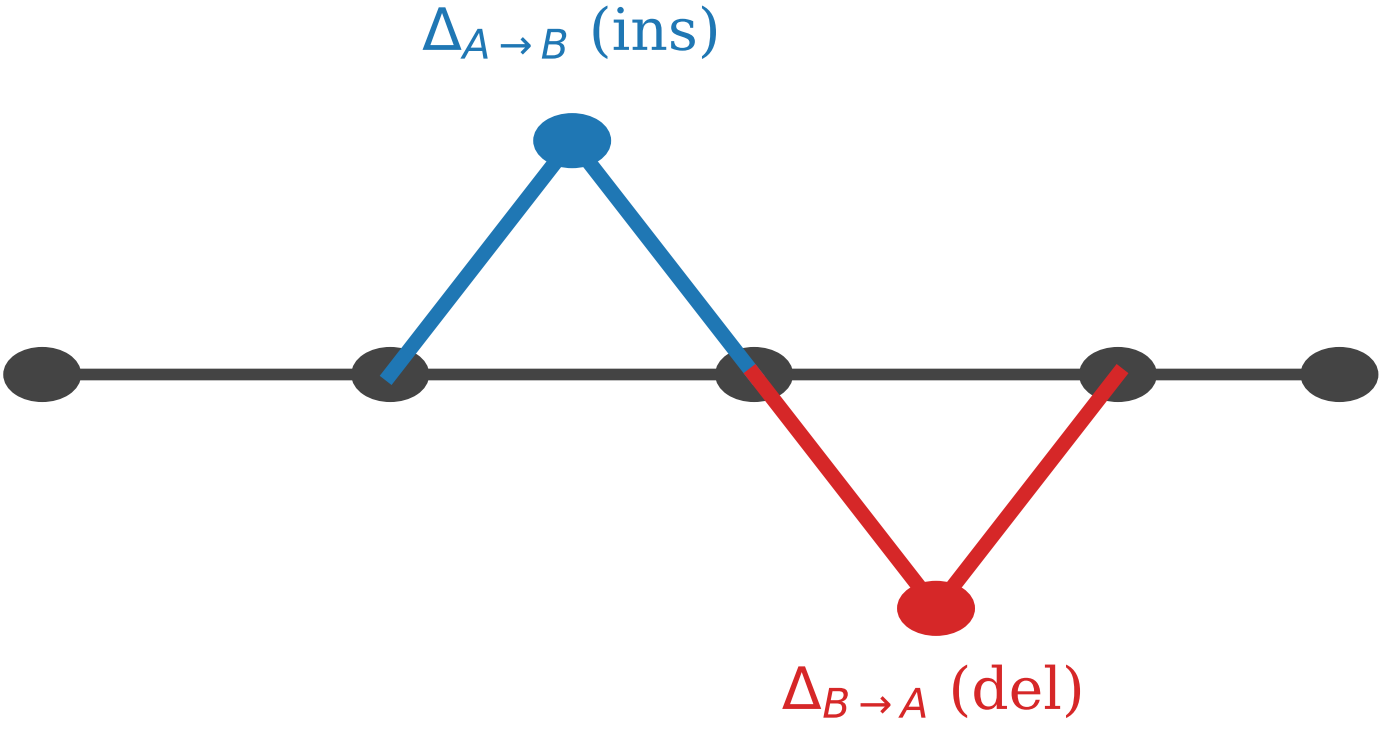shared graph + base-pair deltas